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R V College of Engineering

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

IV Semester Regular/Supplementary Examinations June/July- 2025
Course : CAD/CAM & Robotics-IM343AI

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

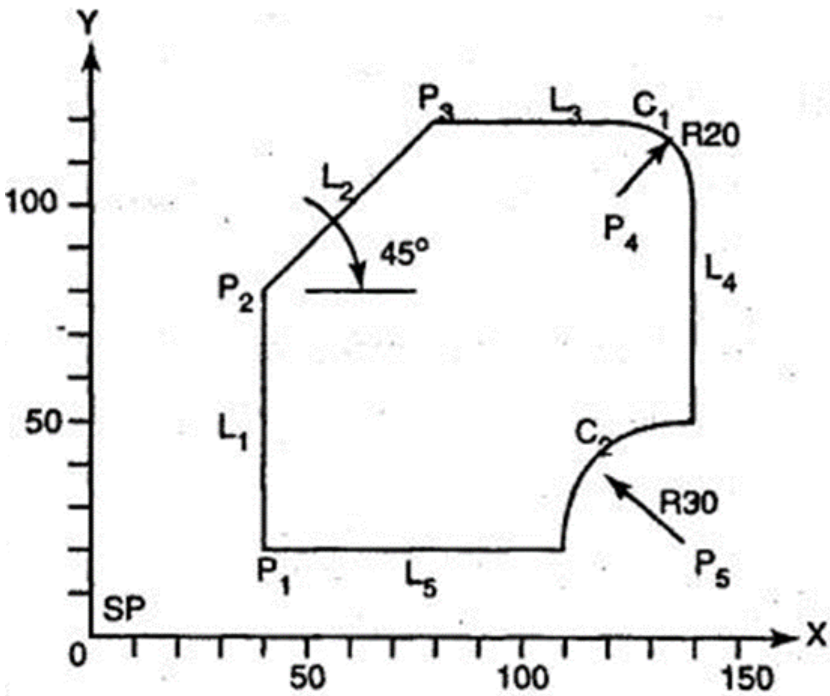
Part A

Question No	Question	M	CO	BT
1.1	Why is conceptual design an important phase in the Shigley model?	02	1	1
1.2	What is the purpose of a line drawing algorithm?	02	1	1
1.3	What is a part program in NC machining?	02	2	2
1.4	What is tool offset in CNC machining?	02	2	2
1.5	What is mechanization?	02	2	2
1.6	List two motivating factors for robot usage in industries	02	3	1
1.7	The power source of a robot can be electric, hydraulic, or ______.	01	3	1
1.8	What is meant by payload in robotics?	02	4	2
1.9	Give two examples of exteroceptive sensors	02	4	1
1.10	What is lead-through programming?	02	4	1
1.11	A __________ is a device at the end of a robot arm, designed to interact with the environment	01	4	1

Part B

Question No	Question	M	CO	BT
2a	Use Bresenham's Line Algorithm to rasterize a line from (10, 15) to (20, 25). Show all intermediate steps.	10	1	4
2b	Discuss the various applications of CAD in different industries **s**uch as mechanical, civil, and electrical engineering.	06	1	3
3a	Analyze the effect of incorrect tool offset values in CNC milling programs.	06	2	1
3b	Compare APT programming with manual NC programming in terms of flexibility, accuracy, and complexity.	10	2	3
OR				
4a	Prepare a program for milling the given component by milling process using APT. (see table and figure below)	16	2	4

Post Processor	Machine Mill, 1
Feed rate	60 mm / min
Spindle speed	500 rpm
Cutter diameter	5 mm
Tolerance	± 0.02 mm
Thickness	10 mm



5a	Explain the social issues associated with automation	08	2	1
5b	Explain reasons for adopting automation in industries.	08	3	3
OR				
6a	Explain how automation contributes to sustainable manufacturing	08	3	4
6b	Explain the different types of automation with examples.	08	2	2
7a	Discuss the selection criteria for industrial robots in manufacturing applications.	08	3	2
7b	Explain the robot wrist and end-of-arm tooling (EOAT) with examples	08	3	2
OR				
8a	A family-owned metal fabrication workshop is evaluating whether to automate its manual cutting and welding operations. However, the owner is hesitant. Analyze the reasons why automation might not be suitable for such a workshop. Provide alternatives.	10	4	4
8b	Explain the classification of robots based on configuration	06	3	1
9a	Describe in detail the role of sensors in robotics and how they improve robot performance.	08	4	3
9b	How do exteroceptive sensors support robotic adaptability in dynamic environments?	08	4	2
OR				
10a	A bakery needs a robot to pick and place soft pastries into boxes. Recommend an end effector and justify it based on hygiene, delicacy, and cycle time.	10	4	5
10b	Explain on-line robot programming and its advantages	06	4	1